

Lionael Surya Dwitama

Toronto, ON | +1 437-606-8490 | suryalionael@gmail.com
linkedin.com/in/lionael-dwitama | github.com/suryalionael | lionael-portfolio.vercel.app

PROFESSIONAL SUMMARY

Data Science and Analytics student (GPA 3.9/4.0) who builds the structured data foundations AI tools depend on. Designed layered PostgreSQL/PostGIS warehouses on municipal open data with decision logs and row-level audit trails, built an LLM service that answers only from curated tables, and shipped the project and documentation platform a nonprofit team runs on daily. Turns messy operational information into structured data and dashboards people can trace to source.

EDUCATION

Seneca Polytechnic

Sep 2024 – Apr 2028 (Expected)

Honours Bachelor of Data Science and Analytics | GPA 3.9/4.0

Toronto, ON

- **Awards:** President's Honour List (Winter 2026); SSF International Student Merit Award
- **Coursework:** Machine Learning, Data Mining, Advanced Database Design, Business Intelligence, Data Visualization

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript, R

Data Architecture: PostgreSQL, PostGIS, star-schema modeling, dbt, Apache Airflow, Kafka, Spark, Delta Lake, DuckDB

AI & Machine Learning: LLM integration (Claude API, OpenAI API), XGBoost, scikit-learn, SHAP, Optuna

APIs, Apps & Tooling: FastAPI, Next.js, React, Supabase, Google Apps Script, Docker, GitHub Actions CI, pytest, Git

Analytics & Dashboards: Power BI (DAX), Streamlit, pandas, GeoPandas

PROJECTS

Toronto Mobility Intelligence | PostgreSQL, PostGIS, Python, Docker, Power BI

Public-Sector Data Platform

- Built a reproducible platform that pulls four City of Toronto Open Data datasets through the CKAN API into a three-layer PostgreSQL warehouse (staging, clean, analytics) with a documented star schema.
- Wrote the Python ETL with schema-drift detection; every excluded or flagged row is logged with a reason rather than silently dropped, so all 20,691 collision records stay auditable.
- Matched 7,587 geocoded collisions to 2,550 signalized intersections with a 20 m PostGIS nearest-neighbour join, choosing the radius from the real distance distribution and storing matches in a bridge table for traceability.
- Kept a decision log; 19 SQL views feed a 4-page Power BI dashboard whose KPIs are reconciled to SQL by 91 pytest tests.

F1 Analytics Platform with AI Race Summaries | Kafka, Spark, Delta Lake, DuckDB, FastAPI, React

Data + AI Engineering

- Built a medallion pipeline streaming OpenF1 API data through Kafka and Spark into Delta Lake Bronze/Silver/Gold tables.
- Integrated a provider-agnostic LLM service (OpenAI or Anthropic) that writes race summaries from Gold analytical tables only, so generated text never reads raw, unvalidated events.
- Served results through DuckDB and a layered FastAPI to React; 48 unit and integration tests run in GitHub Actions CI.

IDX Stock Scanner Agent | Python, XGBoost, Claude API, SQLite, YAML

AI Decision Support

- Built a daily decision-support agent in three layers: a YAML-configured rule engine filters, XGBoost ranks, and the Claude API explains each shortlisted signal in plain language.
- Logged every run's signals to SQLite and backtested them over 3-, 5-, and 10-day windows against real outcomes.

EXPERIENCE

QuickRN Canada

Jul 2025 – Present

Web Development and Data Operations

Remote

- Built Aspen OS solo (Next.js, TypeScript, Supabase), the tasks, kanban, and documentation platform the team uses daily.
- Shipped three production websites, turning non-technical stakeholders' needs into working requirements.
- Managed exam-prep content data and resolved website and course-access issues, documenting each fix for the team.

PERMIKA Toronto

Aug 2025 – Sep 2026

IT and Media Strategy Associate

Toronto, ON

- Built internal tools for event management and member engagement in a volunteer-run organization.
- Led the League of Toronto 2026 social media campaign, coordinating a ten-person team from planning to delivery.

CV. Veda Sakti Dharma

Jun 2025 – Aug 2025

Software Engineering Intern

Jakarta, Indonesia

- Developed and tested a Python application that automates PPh 21 income-tax calculations and payslip generation.
- Organized payroll data for reporting and debugged calculation issues by validating outputs against payroll records.